

Report
Of
Topic: Artificial Intelligence

Submitted To : Mr. Ramavath ganesh

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech 2CSE14



By

Roll Number of Student:

25SCS1003004639

Name of Student:

Shivam kumar

Batch: 2025-29

2026-27

CANDIDATE'S DECLARATION

I, Shivam Kumar do hereby declare that the internship report titled "**Artificial Intelligence**" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copy-right violation in this regard.

Signature : Shivam kumar
Student Name: Shivam Kumar
Roll No.: 25SCS1003004639

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature: Shivam kumar

Student Name :Shivam Kumar

Roll No.: 25SCS1003004639

INTERNSHIP CERTIFICATE



CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Shivam Kumar

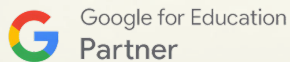
From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Artificial Intelligence Intern

Conducted from 01/06/2026 to 30/06/2026



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, likely belonging to Dr. Anurag Shrivastava.

Dr. Anurag Shrivastava
Program Manager

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INTRODUCTION

Artificial Intelligence (AI) is one of the most rapidly developing areas of modern technology. It focuses on developing computer systems and machines that can perform tasks that normally require human intelligence, such as learning, problem-solving, decision-making, pattern recognition, and data analysis. Artificial Intelligence and Machine Learning are increasingly being used in different fields to automate processes, analyze large amounts of data, and develop intelligent applications.

As part of my academic and professional learning, I completed a **one-month Artificial Intelligence internship at Codec Technologies Pvt. Ltd.** The internship was conducted from 1 **JUNE 2026 to 30 JUNE 2026** and was an **AICTE & ICAC** approved internship program. I worked as an **Artificial Intelligence Intern**, gaining practical exposure to the concepts and applications of Artificial Intelligence.

The main purpose of the internship was to gain practical knowledge beyond classroom learning and understand how Artificial Intelligence concepts can be applied to real-world problems. During this period, I developed a better understanding of the AI development process, including understanding a problem, working with data, applying suitable techniques, developing solutions, and evaluating the results.

The internship also helped me strengthen important technical and professional skills. Working on practical tasks provided an opportunity to improve programming knowledge, logical thinking, analytical ability, problem-solving skills, and understanding of AI-based applications. It also helped me become more familiar with the importance of accuracy, systematic development, testing, and continuous improvement while working on technical projects.

This internship served as an important bridge between theoretical education and practical implementation. The knowledge and experience gained during the program have helped me develop a stronger foundation in Artificial Intelligence and motivated me to explore advanced areas of AI and Machine Learning in the future.

Overall, the internship was a valuable learning experience that enhanced my technical knowledge, practical understanding, and confidence in working with Artificial Intelligence technologies.

Organization Profile:-

Codec Technologies Pvt. Ltd. is the organization where I completed my one-month internship in the field of **Artificial Intelligence**. The internship was conducted from **1 JUNE 2026 to 30 JUNE 2026** as an AICTE & ICAC approved internship program. During the internship, I worked in the role of an Artificial Intelligence Intern.

The organization provided an opportunity to gain practical exposure to Artificial Intelligence and understand the application of technical concepts in a professional environment. The internship environment helped me connect academic knowledge with practical learning and develop a better understanding of AI-related technologies and their applications.

Organization Details

- **Organization Name:** Codec Technologies Pvt. Ltd.
- **Internship Role:** Artificial Intelligence Intern
- **Internship Duration:** 1 JUNE 2026 – 30 JUNE 2026
- **Program:** AICTE & ICAC Approved Internship
- **Duration:** 1 Month
- **Field:** Artificial Intelligence
- **Program Manager:** Dr. Anurag Shrivastava

The organization played an important role in providing practical learning opportunities during the internship. Through the assigned activities and projects, I was able to improve my technical understanding, programming skills, analytical thinking, and problem-solving abilities.

The internship at Codec Technologies Pvt. Ltd. was an important step toward understanding the professional applications of Artificial Intelligence. It provided valuable experience that complemented my academic studies and encouraged me to further develop my knowledge and skills in Artificial Intelligence and Machine Learning.

Problem Statement:-

The rapid growth of digital data and technology has created a strong need for intelligent systems that can process information, identify patterns, and support effective decision-making. Traditional computer programs generally depend on predefined rules and instructions, which can make them less suitable for problems where the data is complex, large, or continuously changing. Artificial Intelligence (AI) and Machine Learning (ML) provide techniques that allow computer systems to learn from data and develop solutions for different real-world problems.

The main problem addressed during the Artificial Intelligence internship was to understand how AI concepts can be transformed from theoretical knowledge into practical, technology-based solutions. The internship focused on developing an understanding of the complete problem solving process, including identifying a problem, understanding the available data, selecting appropriate AI or ML techniques, developing a solution, and evaluating its performance.

Another important challenge was to understand how data can be prepared and used effectively for AI applications. Data may contain inconsistencies, irrelevant information, missing values, or other issues that can affect the performance of an AI system. Therefore, understanding data handling, preprocessing, analysis, and evaluation is an important part of developing reliable AI based solutions.

The internship also aimed to address the gap between academic learning and practical implementation. Although students learn programming, algorithms, and AI concepts through coursework, practical projects provide an opportunity to understand how these concepts are applied to actual problems.

Key Problems Identified

1. Understanding real-world problems that can be solved using Artificial Intelligence.
2. Converting a problem statement into a systematic technical solution.
3. Understanding and preparing data for AI and Machine Learning applications.
4. Selecting appropriate algorithms and techniques according to the problem.
5. Developing and testing AI-based solutions.
6. Evaluating the results and identifying areas for improvement.
7. Bridging the gap between theoretical knowledge and practical implementation.
8. Developing technical, analytical, and problem-solving skills through practical work.

Project Objectives:-

The main objective of this Artificial Intelligence internship project was to develop practical knowledge and technical skills in the field of **Artificial Intelligence (AI) and Machine Learning (ML)**. The internship provided an opportunity to understand how theoretical concepts can be applied to practical problems and how AI-based solutions can be developed systematically. The internship was completed at Codec Technologies Pvt. Ltd. as a one-month Artificial Intelligence internship from **1 JUNE 2026 to 30 JUNE 2026**.

One of the primary objectives was to understand the **fundamentals of Artificial Intelligence** and its importance in modern technology. The project aimed to develop knowledge about intelligent systems and understand how AI can be used for problem-solving, pattern identification, data analysis, and decision-making.

Another important objective was to gain practical understanding of **Machine Learning concepts**. The project focused on understanding how machines can learn from data and how suitable techniques can be applied according to the requirements of a particular problem.

The project also aimed to understand the **importance of data in AI applications**. Proper data handling is essential for obtaining reliable results. Therefore, an objective was to understand basic data preparation, preprocessing, analysis, and the identification of problems such as incomplete or inconsistent data.

A further objective was to **apply theoretical knowledge in practical situations**. Academic studies provide knowledge of programming, algorithms, and AI concepts, while practical implementation helps in understanding their actual applications. The internship provided an opportunity to apply these concepts through practical activities and project-based learning.

The project also aimed to improve **programming and technical skills**. Practical work helped strengthen coding ability, logical thinking, debugging, and understanding of the general AI development process. Along with technical skills, the project focused on developing **analytical and problem-solving abilities** by learning how to divide complex problems into smaller and manageable tasks.

Another objective was to understand the importance of **testing and evaluation**. AI-based solutions need to be tested and evaluated to determine their effectiveness and identify possible improvements. This helped develop a systematic approach toward checking results and improving solutions.

The internship also aimed to **bridge the gap between academic learning and professional experience**. It provided exposure to a professional learning environment and helped develop skills such as time management, task planning, documentation, and systematic working.

Overall, the project objectives were focused on developing a strong foundation in Artificial Intelligence and Machine Learning, improving technical and analytical skills, gaining practical experience, and preparing for future academic and professional opportunities in the field of AI

Scope of the Project :-

The scope of this Artificial Intelligence internship project covers the study, practical understanding, and application of fundamental concepts related to **Artificial Intelligence (AI) and Machine Learning (ML)**. The project provides an opportunity to understand how AI technologies can be used to analyze information, identify patterns, solve problems, and develop intelligent solutions. The internship was completed at **Codec Technologies Pvt. Ltd. as a onemonth AICTE & ICAC approved internship**, conducted from **1 JUNE 2026 to 30 JUNE 2026**, in the role of an **Artificial Intelligence Intern**.

1. Understanding Artificial Intelligence

The project covers the basic concepts and principles of Artificial Intelligence. It focuses on understanding how intelligent systems work and how AI can be applied to different types of technical and real-world problems.

2. Machine Learning

An important part of the project scope is understanding Machine Learning and its relationship with Artificial Intelligence. The project provides a foundation for understanding how computer systems can learn from data and use learned patterns to produce useful result

3. Data Handling and Preprocessing

The project includes an understanding of the importance of data in AI applications. It covers basic concepts related to organizing, preparing, and processing data so that it can be effectively used for developing AI and Machine Learning solutions.

4. Practical Implementation

The scope includes applying theoretical knowledge through practical activities. Programming, logical reasoning, algorithms, and AI concepts can be used to approach technical problems and develop suitable solutions.

5. Problem-Solving

The project focuses on developing a systematic approach to solving AI-related problems. This includes understanding a problem, identifying requirements, analyzing available information, selecting an appropriate approach, implementing

the solution, and reviewing the results.

6. Testing and Evaluation

Testing and evaluation are also included within the scope of the project. A developed solution needs to be checked to determine whether it works correctly and produces meaningful results. Evaluation can help identify errors and possible improvements.

7. Technical Skill Development

The project provides scope for improving technical skills such as programming, data handling, logical thinking, algorithmic understanding, debugging, and basic AI implementation. These skills can be useful for developing future AI and Machine Learning projects.

8. Professional Skill Development

In addition to technical knowledge, the project provides scope for improving professional skills such as time management, task planning, documentation, systematic working, and problemsolving. These skills are important for working effectively in a professional technical environment.

9. Future Expansion

The knowledge gained through the project can provide a foundation for further exploration of advanced areas of Artificial Intelligence and Machine Learning. Future work may include more advanced AI applications and specialized technologies, depending on the project requirements and learning objectives.

Overall Scope

Overall, the scope of the project is to develop a strong practical foundation in Artificial Intelligence and Machine Learning, while improving technical, analytical, and professional skills. The internship experience provides a bridge between academic knowledge and practical application and can serve as a foundation for future AI-related projects, studies, and professional development.

Technologies and Tools Used :-

During the Artificial Intelligence internship, various programming concepts, development tools, and technologies can be used to understand and implement AI and Machine Learning solutions. The main purpose of using these technologies is to simplify data processing, programming, model development, testing, and evaluation. The internship certificate confirms the role as an **Artificial Intelligence Intern**, but it does not specify the exact software, programming languages, or tools used during the internship. Therefore, the following technologies represent the **standard**

tools and technologies relevant to an AI/ML internship, rather than tools explicitly confirmed by the certificate.

1. Python

Python is one of the most widely used programming languages for Artificial Intelligence and Machine Learning. It provides a simple syntax and a large collection of libraries for data analysis, visualization, and machine learning. Python can be used to write programs, process datasets, implement algorithms, and develop AI-based applications.

2. Jupyter Notebook

Jupyter Notebook provides an interactive environment for writing and executing Python code. It is particularly useful for AI and ML projects because code, explanations, results, and visualizations can be maintained together. It also makes experimentation and analysis easier.

3. NumPy

NumPy is a Python library used for numerical computing. It provides support for arrays, matrices, mathematical operations, and numerical calculations. NumPy is useful when working with large amounts of numerical data in Machine Learning applications.

4. Pandas

Pandas is commonly used for data manipulation and analysis. It provides data structures such as DataFrames that make it easier to organize, clean, filter, and analyze datasets. It is especially useful during the data preprocessing stage of an AI project.

5. Matplotlib

Matplotlib is a Python visualization library that can be used to represent data through graphs and charts. Visualization helps in understanding patterns, distributions, relationships, and trends within a dataset.

6. Scikit-learn

Scikit-learn is a popular Machine Learning library in Python. It provides implementations of various machine learning algorithms and tools for preprocessing, model training, prediction, and evaluation. It can be used for classification, regression, clustering, and other machine learning tasks.

7. TensorFlow / Keras

TensorFlow and Keras are widely used for developing deep learning applications. They can be used to create and train neural networks for tasks such as image classification, pattern recognition, and other AI applications.

8. Git and GitHub

Git is a version-control system used to track changes in project files and source code. **GitHub** can be used to store and share project repositories online. These tools are useful for maintaining project versions, organizing code, and collaborating on software projects.

9. Integrated Development Environment

An **Integrated Development Environment (IDE)** or code editor can be used for writing, executing, and debugging programs. Such environments provide features such as syntax highlighting, error detection, project organization, and debugging support.

10. Microsoft Office / Documentation Tools

Documentation tools can be used to prepare project reports, maintain records, organize information, and present the work completed during the internship. Proper documentation is an important part of technical project development.

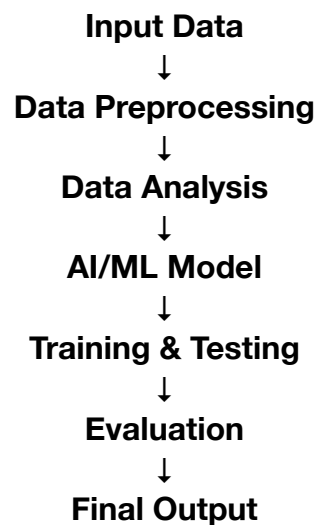
Summary

The combination of programming languages, AI/ML libraries, development environments, dataprocessing tools, visualization tools, and version-control systems provides a complete environment for developing Artificial Intelligence projects. These technologies help in performing different stages of an AI project, including data preparation, analysis, programming, model development, testing, evaluation, documentation, and project management.

System Architecture :-

The system architecture represents the basic workflow of the Artificial Intelligence project. It shows how data moves through different stages to produce the final result.

System Flow:



Description

- **Input Data:** Relevant data is collected for the project.
- **Preprocessing:** The data is cleaned and prepared for further use.
- **AI/ML Model:** Suitable AI or Machine Learning techniques are applied.
- **Training & Testing:** The model is trained and tested using the available data.
- **Evaluation:** The model's performance is checked.
- **Final Output:** The system produces the required prediction, classification, or result.

This architecture provides a simple and systematic workflow for developing an AI-based solution.

Methodology :-

The methodology of the Artificial Intelligence project follows a systematic approach for developing and evaluating an AI-based solution. The main steps are:

1. **Problem Identification:** Understand the problem and define the requirements of the project.

2. **Data Collection:** Collect relevant data required for developing the AI/ML solution.
3. **Data Preprocessing:** Clean and prepare the data by handling missing, duplicate, or irrelevant information.
4. **Data Analysis:** Analyze the prepared data to identify useful patterns and relationships.
5. **Model Selection:** Select a suitable Artificial Intelligence or Machine Learning technique according to the problem.
6. **Model Training:** Train the selected model using the prepared dataset so that it can learn relevant patterns.
7. **Testing and Evaluation:** Test the trained model and evaluate its performance using appropriate measures.
8. **Result Analysis:** Analyze the obtained results and identify possible improvements.
9. **Final Output:** Generate and present the final result produced by the developed AI/ML solution.

Overall Flow:

Problem Identification → Data Collection → Preprocessing → Analysis → Model Selection → Training → Testing → Evaluation → Final Output

Expected Outcomes :-

The Artificial Intelligence internship project is expected to provide practical knowledge and experience in developing AI-based solutions. The major expected outcomes of the project are:

1. **Better Understanding of AI:** Develop a clear understanding of Artificial Intelligence concepts and their practical applications.
2. **Machine Learning Knowledge:** Gain basic knowledge of Machine Learning techniques and understand how models learn from data.
3. **Improved Data Handling Skills:** Learn how to collect, organize, preprocess, and analyze data for AI applications.
4. **Practical Implementation:** Apply theoretical knowledge of programming, algorithms, and AI to practical technical problems.
5. **Improved Programming Skills:** Strengthen programming, logical thinking, debugging, and implementation skills.
6. **Model Development:** Gain an understanding of selecting, training, testing, and evaluating an AI/ML model.
7. **Problem-Solving Skills:** Improve analytical thinking and develop a systematic approach to solving technical problems.
8. **Performance Evaluation:** Understand how to evaluate the results of an AI-based solution and identify areas for improvement.
9. **Professional Skills:** Improve time management, documentation, task planning, and systematic working.
10. **Future Learning:** Build a strong foundation for further studies and projects in Artificial Intelligence and Machine Learning.

Overall Outcome

Overall, the project is expected to strengthen **technical knowledge, practical implementation skills, analytical thinking, and problem-solving abilities**. It will also provide a foundation for developing more advanced AI and Machine Learning projects in the future.

Letter of Recommendation:-



INTERNSHIP OFFER LETTER

Dear Shivam Kumar ,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 01/06/2026 to 30/06/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

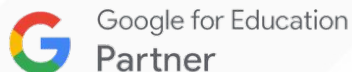
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 01/06/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



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Bibliography/References :-

The following references were used to support the study and preparation of the Artificial Intelligence internship project. These resources provided useful information about Artificial Intelligence, Machine Learning, Python programming, data processing, algorithms, and practical implementation.

1. **Russell, S. J., & Norvig, P.** Artificial Intelligence: A Modern Approach. Pearson Education. This book provides fundamental concepts and principles of Artificial Intelligence, including intelligent agents, problem-solving, learning, and AI applications.
2. **Aurélien Géron.** Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media. This reference provides practical knowledge of Machine Learning, model development, data processing, and neural networks.
3. **Wes McKinney.** Python for Data Analysis. O'Reilly Media. This book is useful for understanding Python-based data manipulation, analysis, and preprocessing using libraries such as Pandas and NumPy.
4. **Python Documentation.** Python Software Foundation. The official documentation was useful for understanding Python programming concepts, syntax, functions, and libraries.
5. **Scikit-learn Documentation.** The official Scikit-learn documentation provides information about Machine Learning algorithms, preprocessing methods, model training, and evaluation techniques.
6. **TensorFlow Documentation.** The official TensorFlow documentation provides guidance on developing and training Machine Learning and deep learning models.
7. **Internship Certificate – Codec Technologies Pvt. Ltd.** The certificate was used to verify the internship organization, internship duration, role, and program details. The certificate records the internship as a one-month **AICTE & ICAC** approved program conducted from **1 JUNE 2026 to 30 JUNE 2026 for the role of Artificial Intelligence Intern.**

These references helped in developing the theoretical understanding and practical foundation required for completing and documenting the Artificial Intelligence internship project.